



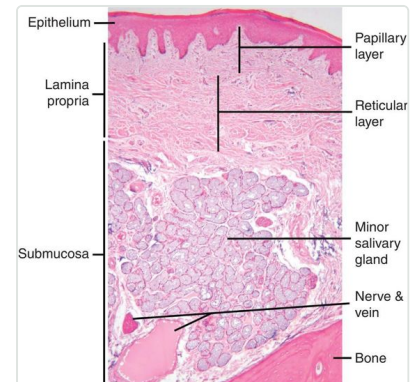
BEFORE YOU READ – 3 BLIND ID CHALLENGES

1. Name the four kinds of lingual papillae — which one has no taste buds, and which detects bitter?
2. Which major salivary gland is largest, which makes ~70% of the saliva, and which is smallest?
3. On a salivary gland slide, what separates an intercalated duct from a striated duct?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 · Oral cavity and lips — one-line ID rules

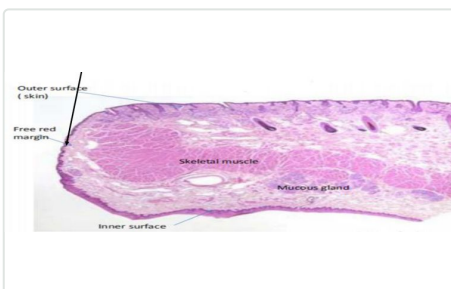
- Oral cavity = first site of **mechanical digestion** and chemical digestion (by saliva); lined by **stratified squamous epithelium** on connective tissue.
- **Cheek:** CT with many **elastic fibres and much fat**. **Hard palate:** mucosa rests on **bone**. **Soft palate:** core of **skeletal muscle** + numerous **mucous glands** in the submucosa.
- Slide layers: epithelium → lamina propria (**papillary** then **reticular** layer) → submucosa with **minor salivary glands**.



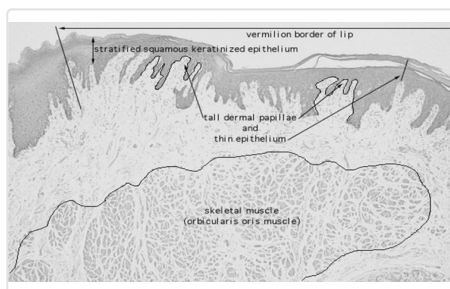
Oral mucosa: epithelium, lamina propria (papillary / reticular), submucosa with minor salivary gland, nerve & vein, bone.

LIP = SKELETAL MUSCLE (ORBICULARIS ORIS, CONCENTRIC ROUND THE ORAL ORIFICE) WITH THREE SURFACES

ZONE	EPITHELIUM	GLANDS / HAIR	ONE-LINE ID
Inner (oral) surface	thick non-keratinized stratified squamous	small clumps of labial (salivary) glands ; sebaceous gland	looks like the rest of the oral mucosa
Vermilion border (transitional zone)	lightly keratinized stratified squamous; tall dermal papillae, thin epithelium	no hair, no sweat, no sebaceous glands	red from prominent blood vessels + many sensory nerve endings
Outer surface	thin skin	hair follicles, sebaceous and sweat glands	skin with adnexa



Lip: outer surface (skin), free red margin, skeletal muscle, mucous gland, inner surface.



Vermilion border: keratinized stratified squamous epithelium, tall dermal papillae, orbicularis oris.



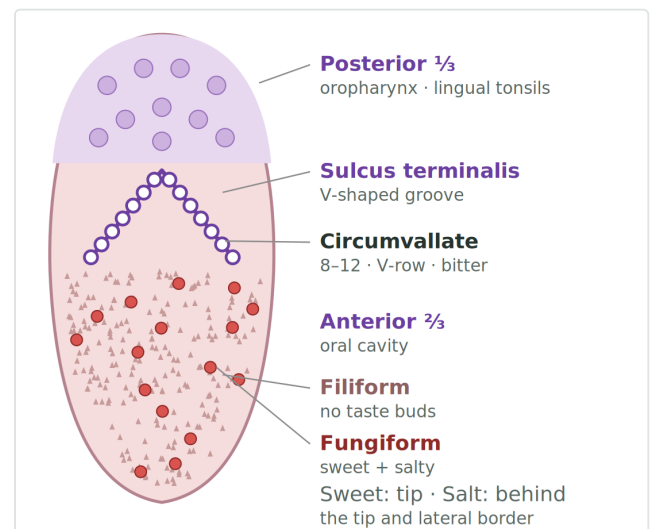
Outer surface with hair follicles, meeting the vermilion zone.

2 · Tongue — muscle, surfaces, papillae, taste buds

One-line ID: mass of **skeletal muscle** in bundles crossing in **3 planes** (flexibility for speech, positioning food, chewing, swallowing), covered by **lightly keratinized** stratified squamous epithelium bound tightly to the muscle by the lamina propria.

SURFACE	EPITHELIUM
Ventral (floor of oral cavity)	non-keratinized stratified squamous
Dorsal (top)	partly keratinized , covered by papillae (epithelium + connective-tissue core)

- **Sulcus terminalis** (V-shaped groove) separates the anterior $\frac{2}{3}$ (oral cavity) from the posterior $\frac{1}{3}$ (oropharynx); **lingual tonsils** lie behind it.
- **Circumvallate** papillae sit in a V-shaped row at the back of the tongue, on the sulcus.
- **Taste buds** sit in circumvallate, fungiform and foliate papillae, and in smaller numbers on the **soft palate, epiglottis, palatoglossal arches, posterior oropharyngeal wall**.
- **Sweet** is best appreciated at the tip; **salt** just behind the tip and along the lateral border; **bitter** by the circumvallate papillae.



Anchor diagram: the dorsal tongue, posterior at the top (as in the lecture figure).

LOOK-ALIKE GRID – THE FOUR PAPILLAE (THE ONE FINDING THAT SEPARATES THEM IN BOLD)

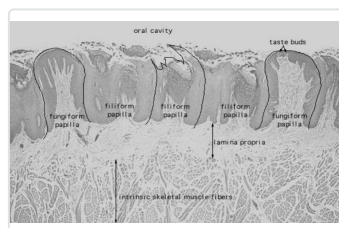
	FILIFORM	FUNGIFORM	CIRCUMVALLATE	FOLIATE
Shape / size	smallest, elongated conical	mushroom-shaped	largest	not described in the lecture
Number / place	most numerous ; whole dorsal surface	occur singly, evenly between the filiform	8-12; V-shaped row at the back	—
Epithelium / core	partly keratinized → roughness, friction to process food	slightly thinner; core richly vascularised	ringed by a groove (infolded epithelium)	—
Taste buds	NONE	scattered · sweet + salty	most numerous on lateral surfaces · bitter	present
Extra	—	—	serous glands of von Ebner open into the groove	—

MAGNIFICATION LADDER (SCAN-PATH ORGANISATION OF THE LECTURE'S FEATURES – THE LECTURE GIVES NO MAGNIFICATIONS)

MAG	WHAT TO LOOK FOR
×4	Tongue: muscle bundles in three planes under the mucosa; serous and mucous glands among the muscle; papillae fringe the surface.
×10	Papilla type: tall pointed keratinized filiform vs. broad fungiform with a vascular core vs. a big papilla in a groove (circumvallate) with von Ebner gland ducts.
×40	Taste bud in the epithelium: taste pore , sensory (gustatory) and supporting cells, basal cells at the base.



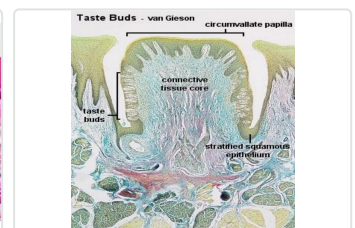
Tongue: oral mucosa, muscle bundles, serous and mucous glands.



Fungiform + filiform papillae; taste buds; lamina propria; intrinsic muscle.



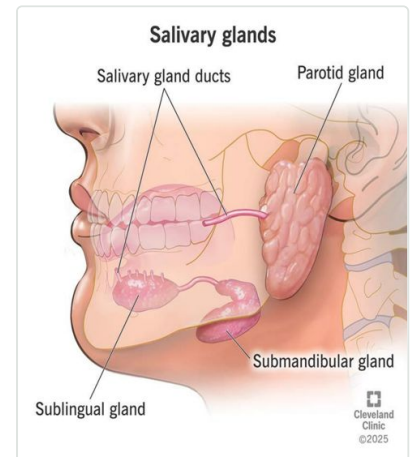
Circumvallate papilla with its groove.



Taste buds, lateral wall of a circumvallate papilla.

3 • Salivary glands

- **Major** (parotid, submandibular, sublingual): each has a **capsule**; septa split it into **lobes** → **lobules** carrying vessels and nerves ("like a branch of grapes"); built of **secretory cells + a duct system**.
- **Minor**: numerous small glands in the mouth (e.g. labial glands of the lip; mucous glands of the soft palate).

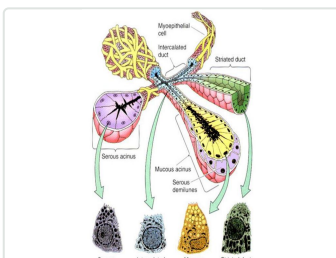


Positions of the three major glands.

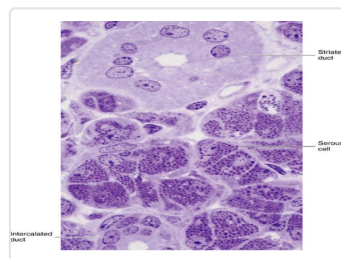
	SEROUS CELLS	MUCOUS CELLS	MYOEPIETHelial
Shape / arrangement	pyramidal ; spherical acinus with a narrow lumen	cuboidal to columnar; arranged as tubules	line the alveoli (listed in the lecture)
Look / product	basophilic cytoplasm, protein secretory granules → protein-rich watery secretion	larger, more acidophilic than serous → mucus	—
DUCT	WHERE	EPITHELIUM / ID	
Intercalated	intralobular (thin CT layer)	narrow lumen , simple cuboidal	
Striated	intralobular	basal striations , simple columnar	
Interlobular	in the septa between lobules / lobes	larger, abundant CT around; simple columnar → stratified	

DISCRIMINATOR GRID – THE THREE MAJOR GLANDS

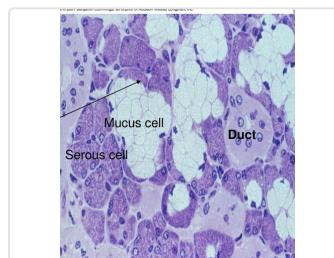
	PAROTID	SUBMANDIBULAR	SUBLINGUAL
Size / place	largest ; beside and in front of the ear	ovoid; neck, just below the mandible	smallest ; floor of mouth, either side of the frenulum
Duct opens	one long duct opposite the 2nd upper molar	floor of mouth, one each side of the frenulum	through 10-12 small ducts
Gland type	compound acinar	compound tubulo-acinar	compound tubulo-acinar
Secretory cells	serous only	serous + mucous, serous predominant ; serous demilunes secrete lysozyme	mucous tubules with serous demilunes
% of saliva	~25-30%	~70%	~5%
Ducts	long intercalated , short striated	long striated , short intercalated	short intercalated; striated ill-defined (slide wording unclear)



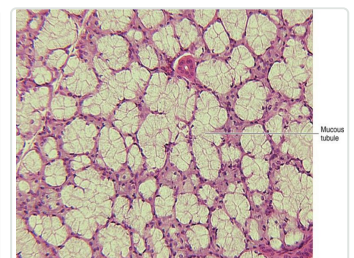
Secretory unit and duct system: serous acinus, mucous acinus, serous demilune, intercalated and striated ducts, myoepithelial cell.



Parotid: serous cells, intercalated and striated ducts.



Submandibular: mucous cells, serous cells, duct.



Sublingual: mucous tubules.

Pearls — what must survive if everything else is forgotten

- **Filiform papillae have NO taste buds** (roughness + friction); fungiform = sweet + salty; circumvallate (8-12, V-row, groove + von Ebner serous glands) = bitter.
- **Sulcus terminalis** = V-shaped groove: anterior $\frac{2}{3}$ in the oral cavity, posterior $\frac{1}{3}$ in the oropharynx, lingual tonsils behind it.
- Lip: inner = thick **non-keratinized** epithelium + labial glands; **vermilion** = lightly keratinized, red, no hair / sweat / sebaceous glands; outer = skin.
- **Parotid** = largest, serous only, ~25-30%, long intercalated + short striated ducts. **Submandibular** = ~70%, serous-predominant mixed, demilunes (lysozyme), long striated + short intercalated. **Sublingual** = smallest, ~5%, mucous + demilunes, 10-12 ducts.
- **Serous** = pyramidal, basophilic, acinus with narrow lumen; **mucous** = tubules, more acidophilic. **Intercalated** = narrow, cuboidal; **striated** = basal striations, columnar.

Test yourself — transfer-task MCQs

Unseen fields, not slide recall. Commit to an answer before you uncover the key.

1. A biopsy from the anterior two-thirds of the dorsal tongue shows numerous small conical projections with partly keratinized epithelium. No taste buds are seen in any section. Which papillae are these?

- A. Fungiform
- B. Circumvallate
- C. Filiform
- D. Foliate
- E. Lingual tonsil crypts

2. A section shows a very large papilla sunk in a groove; ducts of serous glands open into the groove and taste buds crowd its lateral walls. Which taste is this papilla associated with?

- A. Sweet
- B. Bitter
- C. Salty
- D. Sweet and salty
- E. None — it has no taste buds

3. Which finding best separates the parotid from the other two major salivary glands?

- A. Serous demilunes secreting lysozyme
- B. Purely serous secretory cells with long intercalated and short striated ducts
- C. About 70% of the saliva output
- D. Opening into the floor of the mouth beside the frenulum
- E. Ten to twelve small excretory ducts

4. A lip biopsy shows lightly keratinized stratified squamous epithelium with tall dermal papillae and prominent blood vessels. There are no hair follicles, sweat glands or sebaceous glands. Which zone is this?

- A. Inner (oral) surface
- B. Outer skin surface
- C. Vermilion border
- D. Hard palate mucosa
- E. Ventral tongue mucosa

5. A biopsy of the roof of the mouth shows stratified squamous epithelium over a core of skeletal muscle with numerous mucous glands in the submucosa. Which region?

- A. Hard palate
- B. Soft palate
- C. Cheek
- D. Floor of the mouth
- E. Gingiva over bone

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1. **C** — Filiform papillae are the smallest and most numerous, partly keratinized (roughness, friction), and contain no taste buds.
- 2. **B** — Circumvallate papillae (grooves, von Ebner serous glands, lateral taste buds) detect bitter taste.
- 3. **B** — Parotid = only serous cells, long intercalated / short striated ducts, ~25-30%; the other options describe submandibular or sublingual.
- 4. **C** — The vermilion (transitional) zone is lightly keratinized, red from vessels, and lacks hair, sweat and sebaceous glands.
- 5. **B** — Soft palate = skeletal muscle core + mucous glands in the submucosa; the hard palate mucosa rests on bone.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Histology GIT Block · MEDucation by Mattin Majid · Source: GIT Histology SGL 1, Dr Hero K. Mustafa
Study alongside the lecture, not instead of it.